Homework 3: Medicare Advantage Star Ratings & Market Share

Valerie Hernandez

2026-03-06

GitHub Repository: https://github.com/valerie-hdz/homework3

---

1 Question 1 Summary Statistics by Year

The table below reports the mean star rating (among rated plans), mean enrollment, mean market share, and total number of plan-year observations for each year from 2010 to 2015.

Table 1: Summary Statistics for Rated MA Plans, 2010–2015

Year	Mean Star Rating	Mean Enrollment	Mean Market Share	Number of Plans
2010	2.964	252.245	0.065	112,856
2011	3.241	336.362	0.085	70,184
2012	3.329	372.366	0.084	69,770
2013	3.525	384.188	0.080	70,930
2014	3.771	431.075	0.078	65,095
2015	3.940	466.571	0.077	67,505

---

2 Question 2: Plans Without a Star Rating

The tablemsummarizes the mean enrollment, mean market share, and total number of plans for the unrated plans by year.

Table 2: Summary Statistics for Unrated MA Plans, 2010–2015

Year	Mean Enrollment	Mean Market Share	Number of Plans
2010	109.950	0.067	49,536
2011	172.682	0.091	12,763
2012	162.435	0.053	8,743
2013	182.393	0.050	3,246
2014	248.227	0.043	3,324
2015	312.343	0.041	4,729

---

3 Question 3:Distribution of Star Ratings

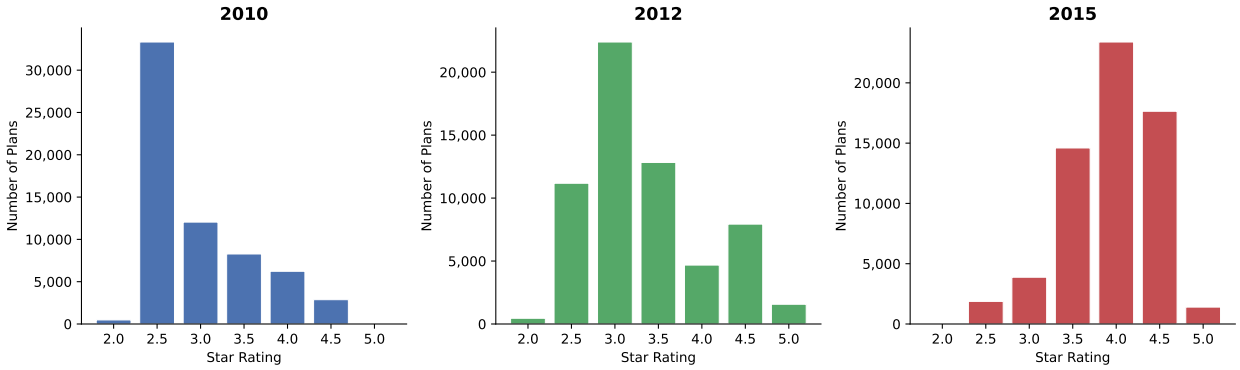

Figure 1: Distribution of Star Ratings for MA Plans (2010, 2012, 2015)

The distributions show how the ratings shifted upward from from 2010 to 2012 to 2015. By 2015, the distribution had shifted toward higher ratings compared to 2010, with fewer plans receiving 2- or 2.5-star ratings and a much larger amount at 3.5 stars and above (right skew for 2010).

---

4 Question 4: OLS Regression of Market Share on Star Ratings

Table 3: OLS Regression of Market Share on Star Rating Indicators (SE in parentheses)

Variable	2010	2011	2012	2013	2014	2015
Intercept (2.5 stars)	0.0623	0.0689	0.0568	0.0427	0.0497	0.0586
	(0.001)	(0.0018)	(0.0021)	(0.0024)	(0.0048)	(0.0038)
3 Stars	0.004	0.0288	0.0549	0.0544	0.0218	0.0202
	(0.0017)	(0.0023)	(0.0025)	(0.0029)	(0.0051)	(0.0046)
3.5 Stars	0.0034	0.022	0.0368	0.046	0.0425	0.0325
	(0.0019)	(0.0027)	(0.0029)	(0.0027)	(0.0049)	(0.0041)
4 Stars	-0.0001	0.0083	0.0038	0.0244	0.0247	0.0144
	(0.0021)	(0.0031)	(0.0036)	(0.0033)	(0.0051)	(0.004)
4.5+ Stars	-0.0053	-0.0068	0.0134	0.0333	0.0241	0.0269
	(0.0027)	(0.003)	(0.003)	(0.003)	(0.0051)	(0.0041)
N	20,409	20,493	22,577	25,531	24,530	23,920
R^2	0.0007	0.0126	0.0294	0.0168	0.0070	0.0050

In 2010, star rating indicators show little relationship with market share, and higher-rated plans (4+ stars) have slightly negative coefficients relative to the omitted group but this changed by 2010-2013.

---

5 Question 5: Running Variable and Plans Near Star Rating Thresholds

Using 2010 data

Table 4: Plans Around Each Star Rating Threshold (2010)

Star Rating	Cutoff	Plans Below Cutoff	Plans Above Cutoff	Total Plans
3-star	2.75	8,681	4,578	13,259
3.5-star	3.25	4,578	3,201	7,779
4-star	3.75	3,864	1,690	5,554
4.5-star	4.25	3,435	194	3,629

Across all thresholds, more plans fall below each cutoff than above it.

5.1 Question 6: RD Estimates at the 3-Star and 3.5-Star Thresholds

Using a regression discontinuity design with a bandwidth of 0.125:

Threshold	Intercept	Rounded	N obs	R^2	RD Estimate	CI Lower	CI Upper
3-star vs 2.5-star	0.012	0.031	5865	0.065	-0.0324	-0.0523	-0.0124
3.5-star vs 3-star	0.100	-0.034	2049	0.014	-0.0711	-0.0964	-0.0458

6 Question 7: Sensitivity to Bandwidth Choice at 0.10, 0.12, 0.13, 0.14, and 0.15

threshold bandwidth	3-star vs 2.5-star	3.5-star vs 3-star
0.100	0.0086	-0.0503
0.120	-0.0227	-0.0825
0.125	-0.0324	-0.0711
0.130	-0.0257	-0.0697
0.140	-0.0039	-0.0598
0.150	0.0732	-0.0852

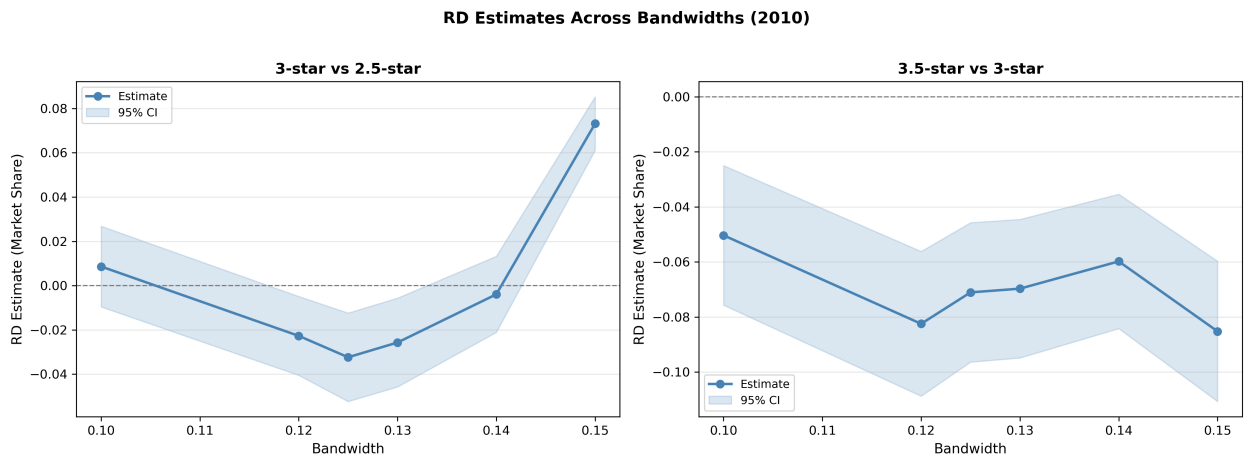

Figure 2: RD Estimates Across Bandwidths

Interpretation:

The 3.5-star threshold estimates are relatively stable across all bandwidths, consistently showing a negative effect on market share ranging from -0.0503 to -0.0852, however, the 3-star threshold estimates are more sensitive to bandwidth choice.

6.1 Question 8: Manipulation Test: Assessing whether Plans Manipulate the Running Variable

```
3-star vs 2.5-star: manipulation test p-value = 0.0000
3.5-star vs 3-star: manipulation test p-value = 0.0002
```

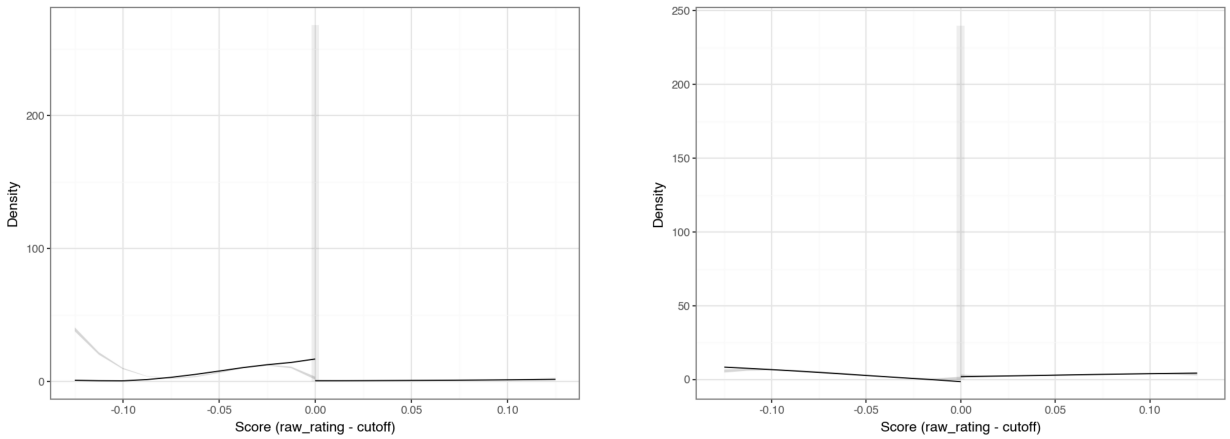

Figure 3: Manipulation Test: Density of Running Variable

6.2 Question 9: Covariate Balance

```
Balance check at 3-star threshold (bandwidth=0.125):

Balance check at 3.5-star threshold (bandwidth=0.125):
Characteristic          Threshold  Coefficient  P-value    N
      HMO Plan 3-star vs 2.5-star       0.6009   0.0000 5865
    Has Part D 3-star vs 2.5-star       0.4309   0.0000 5865
      HMO Plan 3.5-star vs 3-star      -0.5531   0.0000 2049
    Has Part D 3.5-star vs 3-star      -0.0431   0.2025 2049
```

Interpretation: At the 3-star threshold, both HMO status (coef = 0.601, $p < 0.0001$) and Part D coverage (coef = 0.431, $p < 0.0001$) are significant, meaning plans just above the 3-star cutoff are more likely to be HMO plans and to offer Part D coverage than plans just below. At the 3.5-star threshold, HMO status is also highly significant (coef = -0.553, $p < 0.0001$), which could also indicate systematic differences between plans on either side of the cutoff, however, Part D coverage is not significant.

7 Question 10: Summary of RD Findings

```
RD Estimates at Bandwidth = 0.125:
         Threshold  Intercept  Rounded  N obs    R²  RD Estimate  CI Lower  CI Upper
3-star vs 2.5-star      0.012    0.031   5865 0.065      -0.0324   -0.0523   -
0.0124
3.5-star vs 3-star      0.100   -0.034   2049 0.014      -0.0711   -0.0964   -
0.0458

Bandwidth Sensitivity (RD Estimates by Bandwidth):
threshold  3-star vs 2.5-star  3.5-star vs 3-star
bandwidth
0.100                  0.0086             -0.0503
0.120                 -0.0227             -0.0825
0.125                 -0.0324             -0.0711
0.130                 -0.0257             -0.0697
0.140                 -0.0039             -0.0598
0.150                  0.0732             -0.0852

Covariate Balance:
Characteristic          Threshold  Coefficient  P-value
      HMO Plan 3-star vs 2.5-star       0.6009   0.0000
    Has Part D 3-star vs 2.5-star       0.4309   0.0000
      HMO Plan 3.5-star vs 3-star      -0.5531   0.0000
    Has Part D 3.5-star vs 3-star      -0.0431   0.2025
```

Interpretation/Summary: The regression discontinuity estimates suggest that increasing star ratings is associated with lower enrollment at the cutoffs examined, especially at the 3.5-star threshold. And that crossing a star rating threshold does not uniformly increase the market share.